REEFPRINT/KHANYA: belt mineralogy decision support. Analogue-data evidence and a staged PGM validation plan

Mintek–SCi Grad Hackathon 2026 · Problem 3: Computer vision for real-time mineralogical characterisation

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Abstract

South African concentrators set mill feed, grind and reagents before the laboratory can say what was in the ore. REEFPRINT/KHANYA is a research prototype that aims to provide an earlier mineralogical reading for decision support in mineral processing, with PGM concentrators as its intended application; an operator reviews every proposed change and the plant's own controls keep authority. The deployed, authenticated research application runs a pinned model on recorded public copper spectra, binds each prediction to its source and laboratory comparison, and supports inspection of measured hyperspectral images. In a reused 20-record development cohort, a post-selected quartz, biotite and muscovite–sericite panel met a 10% relative-error criterion jointly on 12 records; requiring every eligible phase to meet it gave 0/20. Further public-analogue studies found limited high-abundance subsets with median relative error below 10%, while pretraining and pooling studies did not establish a gain under their stated rules. As an additional processability route, a historical hardness replay reached an out-of-fold R^2 of 0.479 and a simulated 1.7–1.9% throughput gain, but a later evaluation found weaker skill (R^2 0.25) and no combined-gate success. A supplementary microscope prototype maps three sulphides but remains limited by abundance and support failures. None of these endpoints is independent PGM belt validation, and no operational setpoint or realised saving is validated. We propose a bounded host-ore feasibility stage (capped at R210,000), followed, only if its gates pass, by a six-month pilot with supervised shadow operation and matched sensor and laboratory measurements (planning allowance R10.69–13.15 million, proposed to be shared between the Technology Innovation Agency, TIA, and the host). A relative error of ≤10% above agreed reporting limits for PGM belt mineralogy remains a validation target.

1 Introduction

As Team Sonar, working on Mintek's Problem 3, our responsibility is to give a concentrator's shift team a reliable reading of its ore before that ore reaches the mill, and to turn the reading into a plant setting that a person can approve. This report sets out what we built, how we tested it, what it would cost and earn, and, centrally, the six-month pilot that would test it on South African ore (Section 7). It describes a research prototype: the results are on public analogue data and a simulated plant. It is written for metallurgists, mineralogists, process engineers and the people who fund their technology. Technical terms are defined where first used; TRL is the standard technology readiness level, from 1 (basic principles) to 9 (proven in operation) [1].

1.1 Problem statement

The brief states the problem plainly: "without real-time feedback, processing plants cannot adjust to changes in ore quality, leading to inefficient chemical usage and lower mineral yields" [2]. Ore reaches the mill within minutes of leaving the belt, whereas assays and automated mineralogy are batch laboratory work. Mintek's own Minerals Processing and Characterisation Cluster runs QEMSCAN automated mineralogy [3], and a commercial laboratory describes mineralogy as "not an overnight assay service" [4]. Consequently, each shift sets the mill and the flotation circuit for ore whose mineralogy it has not yet seen.

The cost of deciding blind falls most heavily on South Africa, which mined an estimated 120 of the world's 170 tonnes of platinum in 2025 (about 71%) and 70 of 190 tonnes of palladium. It also holds 63,000 of the world's more than 76,000 tonnes of PGM reserves, mostly in the Bushveld Complex [5]. The same source attributes the 9% fall in South African PGM output in 2025 to lower palladium prices, higher deep-level mining costs and disruptions to the electricity supply.

Recovery is where the money is. UG2 ore carries about 30% Cr_2O_3 against about 0.1% for Merensky ore, and the smelter caps the chromite in concentrate. Mintek testwork made UG2 concentrate at 87% recovery with 2.9% Cr_2O_3, and recoveries above 90% were possible only if the Cr_2O_3 limit was relaxed [6]. Producers report recovery moves of this size to investors: Valterra raised concentrator recoveries by 1.2–1.7 percentage points at Mototolo and Amandelbult in 2025 [7]. Plants do not decide entirely blind: operators, advanced process control (APC) and the mine plan already carry information about the ore. Thus, the decision we aim to improve is narrower: setting mill feed for each parcel of ore before it arrives, with a stated uncertainty, at a cost a producer can recover.

1.2 Proposed solution: two measurement cadences, one operator workflow

REEFPRINT/KHANYA reads the ore at two cadences and passes both to one operator workflow (Section 3):

1. **the belt** (REEFPRINT), every 30 seconds: a short-wave infrared (SWIR) camera above the mill-feed conveyor feeds a model that predicts each parcel's grinding hardness with an upper bound, and Bond's law turns the bound into a mill feed-rate proposal. historical recorded-data experiments estimate hardness; the deployed application supports recorded copper inference and measured-image inspection, no conveyor sensor is connected, and the hardness-to-feed-rate path remains proposed;
2. **the laboratory** (KHANYA), once a shift: a microscope model classifies sulphide minerals in polished sections,

Table 1: The brief and the judging criteria: evidence and status. Shown = on public analogue data; Partial = part shown; Proposed = future work; Target = defined, not tested.

Requirement	Evidence (section)	Status
≥ 3 mineral phases	Recorded copper panel and historical microscope labels; no independent three-phase PGM belt validation (4.3, 4.4)	Analogue demo
Accuracy report	Copper 12/20 panel, 0/20 all phases; hardness; microscope by checkpoint (4.1–4.4)	Partial
≤ 10% error	Target defined; untested on independent specimens (2.6)	Target
Adjust a plant parameter	Historical regrind simulation and separate assumption-labelled simulator; belt feed-rate path proposed; no actuation (2.3)	Simulation
Real time	92 ms scoring only; acquisition and belt latency unverified	Partial
Local validation	Six-month pilot (7)	Proposed
Innovation	Bounded input that abstains, operator review (2)	Shown
Feasibility	Pilot, costs, funding, risks, TRL (7–8)	Proposed
Impact	Value per 1% more metal (6.4)	Scenario
Technical execution	Out-of-fold tests, baselines, failures reported (the supporting evidence note)	Shown

which addresses the brief's three-phase requirement on slower, prepared samples;

3. **the decision**: an operator reviews each proposed change before the ore reaches the mill; the site's existing control system keeps authority, and events on the implemented recorded-data route are hashed and logged.

The current application offers recorded copper-spectrum inference with laboratory comparison, measured hyperspectral image inspection, the microscope workbench and an assumption-labelled plant simulator (Section 3); no conveyor sensor is connected, and the 30 s parcel is a proposed cadence. Table 1 maps the brief and the judging criteria published in the hackathon FAQ (2025 edition; we have not seen a 2026 rubric) [8] to the evidence, with the status of each.

2 Methods

2.1 Data

No South African belt scans or labelled Bushveld micrographs were available, so every model was trained and tested on public data; none was supplied by Mintek or a producer. HIDSAG (CC0) supplies visible (39-band) and SWIR (75-band) scans of 146 Chilean porphyry copper–molybdenum drill-core composites with laboratory Bond work-index values [9]. For mineral composition, the same HIDSAG archive supplies recorded mean spectra paired with laboratory mineralogy of copper composites (99 records, 36 composite groups; the deployed 20-record cohort spans seven groups whose physical independence is unresolved), and public paired MLA–hyperspectral sections add area-based phase maps [10]; both are non-PGM, reused in development, and differ in reference basis from PGM belt mass fractions. A set of 1,112 Bushveld chromitite assay intervals (CC BY 4.0) was used to check PGE routing [11]. LumenStone S2 (research use only; 49 micrographs, 37 for training and 12 publisher test sections reused in development), a Norilsk nickel–copper sulphide ore [12], shares the mineral assemblage of Bushveld ores but not their abundance: sulphides cover 62.8% of its sections by area, whereas Merensky ore contains up to about 3% base-metal sulphides and UG2 less [6]. It supplies labels for the relevant phases, because the base-metal sulphides in UG2 are predominantly pentlandite, chalcopyrite, pyrrhotite and pyrite [6]; its much richer sulphide content and different preparation can still cause transfer failure, which only host specimens can test.

2.2 The belt layer: hardness from the infrared

Sensor (design). The design uses a SWIR line scanner such as the Specim SX25 (960–2500 nm, 640 pixels across; 162 lines a second assumed; exposure, band mode and throughput are verified by vendor quote before capacity planning) with a visible–near-infrared (VNIR) companion camera [13]. On a 1.2 m belt at 2 m/s it records a line every 12.3 mm with a 1.88 mm sampling pitch across the belt (not an optical resolution), so it measures averaged parcels of ore, not particles. In the pilot design a *parcel* is 30 s of belt (about 60 m, or 4,860 scan lines averaged into one spectrum); in the replay reported here, a parcel is one HIDSAG composite scan of 80×117 pixels.

Model. Each parcel's spectrum is reduced to features from the infrared absorption regions of clay, mica, chlorite and carbonate minerals (Fig. 2), and a model predicts the Bond work index (W_i, kWh/t). Results are reported only for samples the model never saw (out of fold), and a result counted only if it beat simple baselines under pass marks committed to the repository before each run. Five targets and several model variants were examined and only the work index survived, so the result is exploratory selection, not a confirmatory test; HIDSAG has no drill-hole identifiers, so grouped independence of the folds cannot be verified.

Bound and gate. The plant setting uses an upper bound that the true hardness should stay below 9 times out of 10 (a one-sided 90% split-conformal bound [14]), not the best guess. A more data-efficient bound [15] was tried first and failed its pass mark, so the exact bound that the plan named as its fallback is used, and the result is provisional. The 90% is a marginal coverage target that assumes new parcels resemble the calibration data; it is not a per-parcel guarantee and does not certify protection against overload. An out-of-distribution (OOD) check flags parcels unlike anything in training, and their advice is withdrawn.

2.3 The decision layer and its cadence

The decision layer turns the hardness bound into a mill setting that a person can approve. Ball mills are usually limited by motor power, so the tonnes a mill can grind depend on how hard the ore is. Bond's law gives the specific grinding energy [16]

$$W = 10\, W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \text{ kWh/t}, \tag{1}$$

where F_{80} and P_{80} are the feed and product sizes (μm) passed by 80% of the material. At a fixed grind target the feed rate the mill can carry is proportional to $1/W_i$, so the proposal is

$$F_{\text{proposed}} = F_{\text{design}} \times \frac{W_{i,\text{design}}}{W_{i,\text{bound}}}. \tag{2}$$

The ratio removes explicit power and size terms only when power, feed size, product target and circuit efficiency are held constant; the pilot must test that approximation for the selected circuit. For example, a design hardness of 16.6 kWh/t and a

bound of 16.9 kWh/t give 98.2% of design feed, inside an envelope set by the site's metallurgist (85–110% in the demonstration). The demonstration falls back to 100% of design feed; at a site, the fallback is the host's own operating procedure.

In the proposed cadence a person approves changes, not every computation: under the illustrative 30 s acquisition and 90 s scanner-to-mill transit, a parcel scanned from $t = 0$ to 30 s has its leading material at the mill at $t = 90$ s and its last at $t = 120$ s, so a completed estimate leaves about 60 s; preparation, inference, communication, operator response and actuation must fit inside that, with a margin. Advice expires against the arrival of the material it concerns, and a late estimate is kept only as a retrospective record. These are design assumptions, not achieved latency; belt speed, sensor position, actuator response and acquisition-to-action time (p50/p95/p99, with backlog) are measured at the host. A proposal is computed every 30 s (at most 120 an hour), but the operator sees it only when it differs from the running setpoint by more than a deadband set by the metallurgist (for example 2% of design feed), and may approve, modify within the envelope, reject or escalate. With no approval in time, a flagged parcel or stale data, the advice is withdrawn and the site's own control policy continues; the system never raises feed on its own. Feed may be cut by up to 10% but raised by only 3% per parcel, and the shadow phase measures the real operator-facing rate before anyone acts. The historical decision-record route supports hash-chained events and signed checkpoints (Ed25519 and ML-DSA-65 [17]); signature coverage for each deployed route is recorded in the release manifest, and this supports traceability, not measurement correctness. In the historical experiments the microscope's advice reached a simulated regrind setting over OPC UA [18]; the current method abstains, and the feed-rate path is proposed, not built.

2.4 The laboratory layer: minerals under the microscope

The supplementary microscope model is a standard segmentation network (DeepLabV3, ResNet-50 [19, 20]) that labels each pixel of a reflected-light image as resin, pyrrhotite, chalcopyrite, pentlandite or magnetite, over six 512-pixel fields (*Quick*) or a whole section. Connected regions are split into grains, with chalcopyrite and pentlandite treated as payload and pyrrhotite as reject, a demonstration assumption because pyrrhotite's value depends on PGM deportment. In the historical experiments, a rule-based advisor turned a 2D payload-area proxy (grains at least half payload were "free") into *grind finer*, *continue*, *verify* or *adjust reagent* hypotheses, with a ±0.335 verify band calibrated on the 12 evaluation sections (Section 4.4). The current application (region-aware contract version 3, measurement `observed-target-support-v3`) reports target-area categories instead, records retained target support and each exclusion cause, does not apply the historical margin, and always abstains: operational advice remains withheld pending method validation, applicable calibration, release review and host approval.

2.5 Evaluation method

Belt models were scored out of fold against the strongest simple baseline (the fold mean, the fold median and a colour-only model), with pass marks committed to the repository before each run. The microscope was scored on the publisher's 12 S2 test sections. They were excluded from network training but reused during development, so we report them as *development-influenced evaluation sections*, not an independent test. Accuracy is the intersection over union (IoU: the overlap between predicted and expert-labelled areas, from 0 to 1), pooled over 103.8 million pixels and averaged per section, with bootstrap confidence intervals over sections. Decision-level agreement runs the same advisor on model and expert masks; it measures sensitivity to segmentation error, not whether the rules are right.

2.6 Validation targets and independent protocol (proposed)

The ≤10% error target must be defined before samples are collected, because absolute and relative errors differ: a reference of 2 wt% predicted as 3 wt% is a 1-point absolute error but a 50% relative error. We propose that the mineralogist and a statistician sign an acceptance contract naming the decision-critical phases, an absolute tolerance below an agreed quantification limit and a ≤10% relative target above it, applied per specimen to every decision-critical phase (our proposed definition, Table 7). The contract also fixes the phase panel, reporting floor, coverage (refusals stay in the denominator) and joint endpoint: a median relative error below 10% does not mean 90% of parcels are jointly within 10%, 80% coverage cannot yield 90% success over all attempts, and below-floor absolute tolerance is a separate endpoint. Image-area abundance is judged separately from decision calibration and from physical liberation. Microscope image fractions are 2D areas, whereas QEMSCAN modal results are usually mass fractions, and a density conversion alone does not make a polished surface represent bulk mass. So the comparison unit, aggregation and density assumptions are fixed in advance, and a same-surface area comparison comes before any mass estimate.

A blind test needs more than hiding answers: a separate custodian holds specimen identities and laboratory results; the frozen model receives only acquisition inputs and permitted metadata; predictions are hashed before the reference is revealed; splits are grouped by specimen and collection event; and every refusal stays in the denominator. The 30 development, 10 calibration and at least 30 locked specimens are budget placeholders; final independent group counts and stopping rules follow the signed precision and power design, calibration fitting is kept separate from release evaluation, and a failed model faces a new untouched cohort. The protocol is drafted, not signed (Section 7.2).

3 System architecture

Figure 1 illustrates the system, highlighting that both branches share provenance and operator controls but have their own quality checks. The belt and laboratory layers run on different time scales: laboratory results give slower context and audit evidence, not a check on each 30 s parcel. The historical belt replay contains hardness estimates and decision-record experiments; the deployed routes are recorded copper inference, measured hyperspectral inspection, the microscope workbench and a separate process simulator, each bound to the release manifest. The microscope workbench (KHANYA) is a web application (React) on a Python service (FastAPI and PyTorch) that runs on an ordinary laptop processor. In local mode it accepts connections only from the same machine and makes no external requests, so local image analysis and evidence review can continue offline in the verified local config-

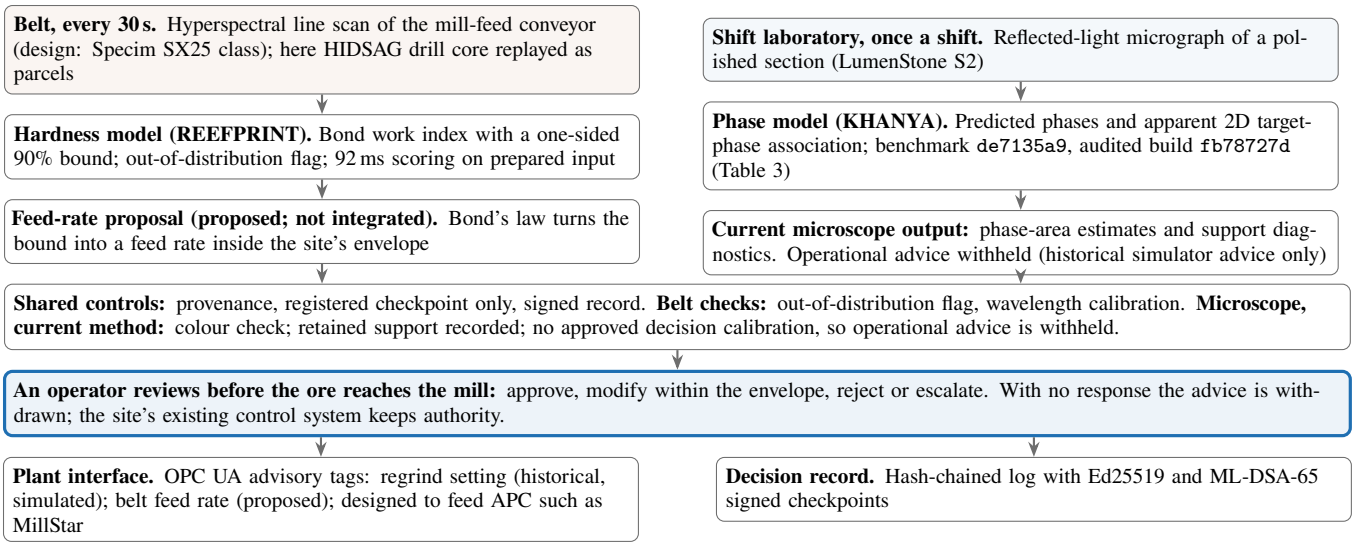


Figure 1: System architecture: two measurement cadences, one operator workflow. The belt and microscope branches produce model estimates at different cadences; the microscope gives predicted phase labels and apparent 2D target-phase association, and independent laboratory measurements provide the reference evidence. Shared provenance and operator controls are supplemented by branch-specific checks. Current deployment: recorded copper inference; measured hyperspectral viewer (no compatible phase model); microscope research output; separate simulator. Simulated (historical): the plant and regrind interface. Proposed: the belt sensor and feed-rate path. Microscope advice is withheld in the current method.

uration. No language model computes a measured value or a control value. Each result records the input and checkpoint hashes, a versioned measurement recipe, the sampled-field coordinates and the share of the image actually observed, and is refused if any of these goes stale, so every number can be traced to what produced it.

The historical 92 ms median is a Kaggle CPU scoring benchmark on already prepared input, not camera-to-decision latency, which is unverified. Microscope timings for the benchmark checkpoint are log-reported, without retained raw records: 16.7–21.2 s per six-field scan on mains power (5 runs) and 30.8 s on battery; a fresh six-field scan in the audited build (`fb78727d`) took about 81 s on the audit machine, so the release chosen for the pilot will be re-benchmarked. The prototype is now an authenticated CPU application on Azure Container Apps; deployment records identify revision `reefprint-judges--0000008`, a frontend-only overlay on the revision-7 backend and model, whose authenticated live acceptance is pending. Recorded copper inference and laboratory comparison are saved with source and model identity; edited predictions are recomputed or refused. The image workspace imports measured hyperspectral cubes, links selected pixels to their spectra and replays measured rows (a display rate, not a sensor line rate); RGB context photographs are not registered to the spectra, and with no compatible deployed model the viewer issues no phase map or plant recommendation. Process calculations are a separate, assumption-labelled simulator with no plant command. Personal workspaces are owner-scoped; team role-based access is not implemented. Inference, start-up, network and acquisition latency are separate, and only scoring time is benchmarked. These checks establish bounded software functionality, not PGM accuracy or live conveyor performance (supporting evidence note).

4 Results

4.1 Belt: predicting grinding hardness

In the historical evaluation design, of the five HIDSAG targets tested (copper and molybdenum recovery, lime consumption, pH and Bond work index), only the work index beat its strongest baseline under every pass mark. Out of fold, the hyperspectral model explains 48% of the variance in work index (R^2 0.479, mean absolute error 1.05 kWh/t), a model given only colour explains 21% (R^2 0.210, 1.29 kWh/t), and the fold mean and median explain nothing (R^2 −0.027 and −0.041). Mean measured W_i is 14.62 kWh/t. A later, differently designed evaluation (EXP23, Table 2) found weaker skill (mean absolute error 1.26 against a 1.45 kWh/t baseline; R^2 0.25) that failed its combined significance gate, so both results are reported.

Figure 3 illustrates how the bound behaves, highlighting that it was exceeded on 10 of 146 samples (6.8%). The 95% interval for coverage (88.4–96.6%) spans the 90% target, so the result is compatible with it but does not establish future coverage, particularly on new ore. The price of the bound is width: the bound sits on average 1.64 kWh/t, about 11% of the mean hardness, above the best guess. The OOD check passed 137 parcels, marked 5 borderline and refused 4. For parcel GMET-0004 (Fig. 2) the bound of 16.9 kWh/t against a design hardness of 16.6 kWh/t (the training data's 90th percentile) gives 98.2% of design feed; the laboratory value was 12.1 kWh/t, so the bound was conservative here.

4.2 Belt: from hardness to a plant decision

Four feed-rate policies were replayed over all 146 composites against a fixed design-feed policy (the 90th-percentile hardness), to show the trade-off between throughput and overloads (simulated; not PGM ore, not a time series). In the replay a parcel is *overloaded* when its true hardness exceeds the hardness the feed was set for; this is a model quantity, not a plant trip. Setting the feed from the bound, with OOD parcels withdrawn, gave a simulated throughput gain of +1.9% (95% CI +0.9 to

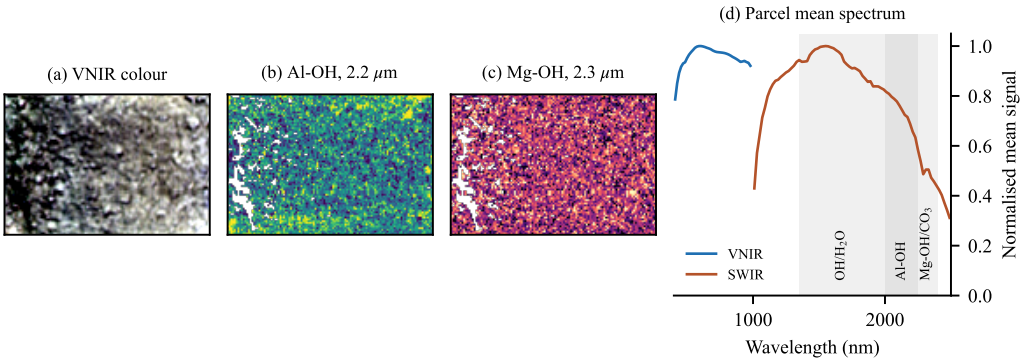


Figure 2: One belt parcel as the hardness model sees it: HIDSAG composite GMET-0004 (Chilean porphyry Cu–Mo), 80×117 pixels. (a) Colour image from the visible bands. (b, c) Relative absorption depth at the Al–OH (white mica, clays) and Mg–OH (chlorite, biotite, carbonate) features; detector stripes removed for display only. (d) Mean spectrum, with the absorption regions the model uses. The model predicts W_i 15.9 kWh/t with a 90% upper bound of 16.9 kWh/t (laboratory value 12.1 kWh/t). The maps show absorption features, not a mineral identification.

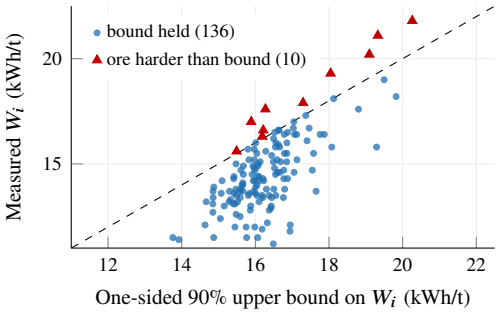


Figure 3: Calibration of the hardness bound, out of fold, 146 composites (no drill-hole grouping available). Points below the dashed line are samples the bound covered: 93.2% (95% CI 88.4–96.6%) against the 90% target. The bound sits on average 1.64 kWh/t (range 1.04–2.14) above the prediction.

+2.9%), and the overload share was 6.8% against 8.9%; the one-sided 95% upper bound on the difference (+1.4 points) is inside the +3-point margin in the evaluation plan. The original replay did not apply the 85–110% envelope; a later analysis with it, reproduced from pinned inputs (the supporting evidence note), gives +1.7% (+0.8 to +2.7%) and the same overload share. Feeding from the best guess instead would add 13.6% but overload 45.2% of parcels, which is why the policy uses a bound. Perfect knowledge would add 14.0%; the bound's width is where the rest is lost, so a tighter bound from site data is the main route to a larger gain. In the evaluated flotation scenarios grinding faster cost up to 0.25 percentage points of recovery on the steep part of the recovery curve, the largest modelled loss rather than a physical bound, so the site's curve must be measured.

The result is provisional for three reasons: the exact bound was adopted after the more efficient one failed, HIDSAG has no time order (so transit time, stockpile mixing and the asymmetric ramp limit are not simulated), and the ore is not Bushveld ore. Over 200 random parcel orders, a ±10% ramp limit left overloads unchanged but ±2% added 4.7 points, so feed must be cut fast. A predefined robustness test found the error unchanged for a new capture, half the field of view, added noise and ±15% illumination, but 1.25 times larger, unflagged, under wavelength drift, so a wavelength-calibration check was added.

Table 2: Mineral-composition development results (relative error above an exploratory 1 wt% floor where stated). EXP21–23 from inspected developer reports; EXP25 recomputed by the coordinator; EXP26 developer report with developer recomputation. Results across the reported runs and controls differed (kaolinite medians about 16.7–19.3%); future comparisons report repeated-seed uncertainty within a fixed recipe separately from design changes.

Study	Result	Interpretation
Deployed copper, quartz–biotite–muscovite panel	12/20 jointly within 10%	Post-selected, reused; 7 composite groups
Same 20 records, all eligible phases	0/20	Broader endpoint fails
EXP21 kaolinite (Li-clay), all	Median 16.3%	Non-PGM development
EXP21 high-prediction subset	Median 7.3%; 84/129 within 10%	About 18% of 703 issued
EXP22 dolomite, infrared	Median 11.4%; selected 5.9% (6/9)	Area share; small subset
EXP22 quartz subset	Median 9%; 5/9 within 10%	Very small, selected
EXP23 copper recovery	96.6% within 10%	Constant baseline 93.8%: no skill shown
EXP23 Bond work index	MAE 1.26 kWh/t, R^2 0.25	Baseline 1.45; failed combined gate
EXP25 pretraining	Kaolinite 19.1% vs 19.3% scratch, 18.5% larger control	No established gain under this recipe
EXP26 pooling	All adjusted $p = 1$	No effect for this recipe

4.3 Belt mineral composition: development results

Table 2 gathers the mineral-composition development results with their denominators. The deployed copper panel's 12/20 is a post-selected result on a reused cohort and sits beside the 0/20 for all eligible phases; the high-abundance subsets below 10% median error cover only part of the issued cases; and the pretraining and pooling studies did not establish a gain. None is independent PGM belt validation.

4.4 Laboratory: supplementary microscope prototype

The microscope demonstrates labels for three relevant sulphides on analogue prepared sections; it does not satisfy

Table 3: Model registry. Scores are bound to one checkpoint each. As of 7 October 2026, no checkpoint has been frozen for an independent test.

Checkpoint	Evidence	Status
`de7135a9`	Mean IoU 0.573 pooled, 12 development-influenced sections	Historical evaluated; research demo
`fb78727d`	Mean IoU 0.454 (re-run); audit on test_11: pixel accuracy 64.1%, pentlandite 34.8% vs 2.0%	Active in audited build; not approved
`42646cfa`	Mean IoU 0.632; pentlandite IoU 0.446, precision 0.50	Candidate; quarantined

an independent PGM belt abundance endpoint. The historically evaluated checkpoint (`de7135a9`) reached IoU 0.87 for pyrrhotite, 0.58 for chalcopyrite and 0.55 for pentlandite on 12 publisher sections reused in development (pooled mean 0.573, 95% CI 0.494–0.624; per section 0.467; majority-class baseline 0.117, colour-only 0.395); magnetite, 0.8% of pixels, is never predicted. Three checkpoints must not share scores (Table 3): the audited build ran `fb78727d`, marked not approved, which on one development section predicted pentlandite at 34.8% of the area against 2.0% in the expert mask, a concrete counterexample to any blanket ≤10% claim; the retrained candidate is quarantined, and two further continuations of it were rejected after raising pentlandite abundance error from 2.2 to 5.4 and 4.4 points.

Historical advice from the model's masks never disagreed with the expert masks in the unsafe direction, but gave only 4 definite actions on 12 sections, with a verify band calibrated on the same sections; 0 of 4 bounds the failure rate only at 52.7% (one-sided 95%), and an exposure shift changed the advice on 8 of 12 sections. The current application excludes edge-connected and overlapping components and abstains: even the expert reference masks keep only 0–5.8% of target pixels in Quick mode and 0–70.5% in full mode, below the provisional 80% support policy, which points to the component-as-particle assumption and specimen preparation rather than segmentation alone.

5 Comparison with literature and limitations

5.1 Comparison with literature

Published work supports the plausibility of the approach, not its validity here. Notole et al. [21] validated hyperspectral imaging and portable XRF against reference methods on Merensky Reef drill core scanned at the Council for Geoscience in Pretoria, with long-wave infrared as well as VNIR/SWIR sensing; that shows the sensing works on Bushveld material, not that our VNIR/SWIR hardness model does, and it is not a public paired conveyor dataset. The three sulphides the microscope classifies are the main base-metal sulphides of UG2 ore [6], and our 50% proxy line coincides with a middling-class boundary in Mintek's liberation scheme [3].

Several ideas were tested and withdrawn: on copper data the belt spectra did not beat a process-line or metadata baseline for chalcopyrite and pyrite (B02 and Table 2), so belt composition remains a development and pilot question; belt chemistry routed Bushveld PGE ore worse than the mine plan's seam label (balanced accuracy 0.79 against 0.82); and the belt did not predict recovery, lime or pH better than a baseline.

5.2 Limitations

The following limitations apply, and (a), (b), (e) and (h) block deployment and performance commitments until tested, and justify a separately scoped feasibility investigation: (a) no South African ore has been evaluated, and models trained on Chilean core may not transfer; (b) the hardness result is a replay of 146 composites without time order or drill-hole grouping, against a fixed policy rather than existing control, and a camera sees only the belt surface; (c) belt mineralogy remains exploratory public-analogue development (copper and paired MLA–hyperspectral data), with no qualified moving-PGM-ore validation, and the microscope has not been trained on chromite or the silicates; (d) the 12 microscope sections guided development, and the historical checkpoint cannot be regenerated exactly; (e) the historical advisor's rules are unvalidated, and payload-dominance is a 2D proxy, not liberation; (f) only scoring latency is benchmarked and no benefit has been attributed at any site; (g) the active checkpoint differs from the evaluated one and reference uncertainty is unquantified; and (h) inadequate support even in reference masks leaves specimen preparation and calibration unresolved.

6 Business model and financial case

6.1 Value proposition and business model

Our value proposition is simple: the plant gets a reading of its ore before the mill, with a stated bound, and beyond a fixed support fee pays a success fee only on independently measured value. No producer, Mintek or TIA has agreed to fund, host or license the work. The buyer is the concentrator or metallurgy manager at a South African PGM producer; named producers are market segments, not customers. TIA [22] and the host would fund the pilot at cost (Section 7.5); reference work would be procured from an agreed laboratory, and any Mintek role and IP route through its Office of Technology Transfer (MOTT) remain subject to contract; an independent auditor would verify the value on which a success fee, capped at 20%, is charged.

The commercial structure has a capped validation contract, a service fee after acceptance and an optional fee capped at 20% of verified net value, zero when that value is zero or negative, with baseline, period, adjustments and audit rights set in advance. Willingness to pay is untested and laboratory savings are not claimed. Mintek's benefit-sharing model with Hindustan Zinc has generated over R35 million since 2021 [23]. Plants already have cross-belt analysers (licensed sources [24]), slurry analysers, automated mineralogy and APC such as MillStar; the system replaces none of them, and the host trial must test incremental value against existing control and blending practice, not only a fixed replay policy.

6.2 Pilot budget and cash release

Table 5 sets out one scoped planning allowance for the six-month pilot: R6.59 million for the laboratory, team and reference track and R4.10–6.56 million for the belt track (the earlier R8.2–49.2 million installation grid survives only as a superseded sensitivity). Staff rates are team planning rates; 44 microscope analyses (40 distinct specimens plus four repeats)

Table 4: Cash release by track (planning allowances; payer agreed at G0). Rows are rounded; the total is summed from unrounded values.

Tranche	Released at	Pays for	Rand
P0	Before G0	Feasibility stage (deliverables in text)	≤0.21 M; outside total
L1	G0	Staff and site, months 1–2; kit; inductions; first sections; 20 bench Bond and mineralogy references	1.75 M
L2	Week-8 laboratory review (not G1B)	Remaining staff, site, sections, microscope mineralogy, review	2.72 M
R-H	G1B pass	24 shadow-composite Bond references	0.47 M
R-M	G1M pass	24 shadow-composite mineralogy references	0.79 M
B1	G0	Sensor lease deposit or bench hire; edge computer	0.57–0.90 M
B2	Bench matrix, week 8	Installation, rest of sensor, belt sampling, vendor support	2.71–4.35 M
Reserves	Documented risk event	15% laboratory, 25% belt contingency	1.68–2.17 M
Total			10.69–13.15 M

and the 44 belt composites go to automated mineralogy, because Bond tests cannot validate mineral composition; and every line is replaced by a local quote at G0. The sensor is leased with a purchase option where possible, so failing both bench gates returns it before installation. VAT, duties, civil work, shutdown costs and closed-loop control are excluded, so totals are scoped budgets, not worst-case losses; post-pilot support is assumed at US$30,000–60,000 a year.

The tracks have independent tranches (Table 4), so a belt failure cannot stop the laboratory work. The 20 bench composites' Bond and mineralogy references (about R1.05 million) are in L1; the 24 shadow composites' references are ring-fenced as R-H (Bond, R0.47 million) and R-M (mineralogy, R0.79 million), each released only if its endpoint passes its bench gate. If both fail, B2, R-H, R-M and the belt reserve are never released; B1, lease cancellation and paid bench references are committed. The pre-G0 feasibility stage (P0) is the first funding request: capped at R210,000 over about three months, proposed for TIA or host funding and approved jointly by co-leads and funder (host and stream scoping R40k; measurement and decision contract R30k; mass and split plan R60k; quotes R20k, Hoaeane; eligibility, IP and data terms R25k, Khumalo; reserve R35k). If host access or an eligible funding route is not secured within the cap, P0 stops and reports; with P0 fully spent, the combined envelope is about R10.90–13.36 million. The 77 sections include 7 preparation duplicates; whether the US$2,000 anchor covers belt-composite crushing and sectioning awaits vendor scope. The cost ledger (evidence package) gives each line an ID, tranche, payer, release, endpoint and cancellation exposure; belt-mineral references sit in the laboratory track for cash control, but their eligibility follows the belt component and is unconfirmed.

6.3 From throughput to recovered metal

The belt result is a throughput gain, and revenue comes from recovered metal, so the two must be bridged. For a consistent stream, metal output is proportional to dry tonnes × head grade × recovery, so

$$\frac{\Delta M}{M} = \left(1 + \frac{\Delta T}{T}\right)\left(1 + \frac{\Delta G}{G}\right)\left(1 + \frac{\Delta R}{R}\right) - 1, \tag{3}$$

where T is throughput, G head grade and R recovery. With unchanged grade, a 1.75% throughput gain and a recovery loss of 0.25 percentage points from 87% (a relative change of −0.29%) give 1.46% more metal. The affected-stream fraction f is unknown until a host is named. Where the plant is limited by ore supply (f effectively 0 for throughput), extra throughput is worth nothing, and a recovery loss larger than the gain would make value negative. Only as an upper-bound scenario, if all of Zondereinde's output passed the instrumented circuit ($f = 1$), 1.46% more metal would be worth R64–123 million a year in margin, and a quarter of that gain R16–31 million; these are simulated gains, not measurements.

6.4 Financial methodology

The following assumptions apply: (a) annual output is that of Northam's Zondereinde mine, 333,050 equivalent refined 4E ounces in F2026 [28]; (b) prices are Valterra's realised basket prices of R32,611 per ounce (2025) and R45,993 (first half of 2026), with cash operating costs of R19,488 and R20,677 per ounce [7, 29], used as proxies for another company's mine, with different baskets, payability and cost structure; (c) for a commercial-life case, installed cost is the belt allowance without the six-month support line (US$175,000–280,000, R2.87–4.59 million) and running cost is the post-trial support (R0.49–0.98 million a year); (d) the equipment has a 5-year life at a 10% discount rate; and (e) an affected-output fraction f of the mine's output passes the instrumented circuit, unknown; $f = 1$ is shown only as an upper bound and $f = 0.25$ as an illustration. The annual value of an uplift u in recovered metal is

$$V = u \times f \times Q \times m, \tag{4}$$

where Q is annual output in ounces and m the margin per ounce. For u = 1% and m = price less cash cost,

$$V = 0.01 \times 333{,}050 \times (\text{R13,123 to R25,316}), \tag{5}$$

which is equivalent to R43.7–84.3 million a year (R108.6–153.2 million in revenue before costs). Deducting the full average cash cost may be conservative where the mill has spare capacity, but extra mining, handling or bottleneck costs could change that, so a host-approved incremental cash-flow model is needed before investment. The installed cost is spread over the equipment's life with the capital recovery factor $\text{CRF} = r(1+r)^n/[(1+r)^n - 1] = 0.2638$ (r = 10%, n = 5), so the annualised cost $C = \text{CRF} \times C_{\text{installed}} + C_{\text{running}}$, which is equivalent to R1.25–2.20 million a year, or R2.50 million with the 25% contingency on the high case. The break-even uplift is then $u_{\text{BE}} = C/(f \times Q \times m)$. The more recovered metal needed is 0.015% at low cost and 2026 margin with all output affected; 0.050% at high cost and 2025 margin; 0.063% if the 20% fee is charged before system costs; 0.20% if only a quarter of output is affected or the basket price falls 30%; and 0.52% on the superseded R49.2 million installation grid. A 20% fee on gross uplift margin (break-even when $0.8\,V = C$) differs from 20% of independently verified net value after system costs; the main case is chosen once the contract is agreed, and line-level costs carry IDs in the ledger. The break-even figures are far smaller than the precision with which a short plant trial can measure uplift, which is why measured value, not break-even, decides G5.

Table 5: One scoped budget for the six-month pilot: team planning allowances, not quotations. US$ list prices are public anchors from one university facility and one industry guide; rand at R16.4 per US$ (January–September 2026 average [25]). Every line is replaced by a quote at G0.

Laboratory, team and reference track			**Belt track: one belt on site**	
Item	Rand	Basis	Item	US$ (rand)
Two pilot engineers (the authors)	480,000	2 × 6 months × R40,000	Sensor, optics, references (lease or purchase-equivalent)	100–160k (R1.64–2.62 M)
Software and OT-security engineer	360,000	To recruit; 6 months × R60,000	Lighting, enclosure, mounting	25–40k (R0.41–0.66 M)
Independent mineralogist	280,000	35 days × R8,000; annotates development specimens	Edge computer, UPS, network	10–15k (R0.16–0.25 M)
Statistician and data custodian	144,000	18 days × R8,000	Installation, integration, commissioning	40–65k (R0.66–1.07 M)
Site living, travel, medicals, inductions	318,000	3 people × 6 months; PPE	Belt sampling crew, references, vendor support	25–40k (R0.41–0.66 M)
Polished-section preparation	154,000	77 sections × R2,000	25% contingency	50–80k (R0.82–1.31 M)
Automated mineralogy, microscope	1,443,000	44 analyses (10 cal. + 30 locked + 4 repeats) × US$2,000 [26]	**Belt track total**	**250–400k (R4.10–6.56 M)**
Automated mineralogy, belt composites	1,443,000	20 bench + 24 shadow × US$2,000		
Bond ball-mill work index	866,000	20 bench + 24 shadow × up to US$1,200 [27]		
Microscope imaging kit	120,000	Allowance		
Independent review, IP and data terms, close-out	120,000	Allowance		
Contingency (15%)	859,000	Released only on a documented risk event		
Laboratory, team, reference total	**6,587,000**	No overrun without a new decision	**Pilot total, both tracks**	**R10.69–13.15 M**

The 1.7–1.9% is a simulated throughput gain, not a realised mine gain. Hosting the research application (one CPU replica, scaling to zero) is a small running cost separate from pilot capital; an 11 October Azure calculation gives US$9.32 and US$13.64 a month for 20 and 40 active hours plus a basic registry, excluding other charges (a scenario, not a quote, cap or bill), and hosting continuity after the current credit is not yet decided. For the pilot itself, R10.69–13.15 million is a scoped planning allowance, not a maximum exposure, because VAT, duties, shutdown and civil costs are excluded and must be assigned to a payer. Tranches (Table 4) leave later money unspent at any stop; the belt range is an allowance pending a lease quote and cancellation schedule, since purchase-equivalent economics do not set lease cash payable. The fee basis must be agreed so that no support cost is charged twice, and measured value needs an agreed metallurgical accounting boundary and attribution method under IPMVP [30].

7 The six-month pilot

About three months of P0 precede this programme, which runs from readiness; access, contracting and delivery may extend elapsed time (roughly nine months in total, not a deadline). As Team Sonar, our pilot problem is this: in six months at one host concentrator, find out whether the microscope meets a ≤10% phase-abundance target on independent local specimens, whether parcel-level belt estimates of mineral composition can meet the same target against matched laboratory mineralogy, and whether the hardness bound holds on the host's own belt, in shadow and then in a short supervised advisory window. Both layers run on site. A clear negative answer at any gate is a useful result, because it stops spending early.

7.1 Pilot duration and feasibility

The following assumptions make six months realistic: (a) contracting, P0 findings, a leased sensor with a delivery date and a plant stop booked for weeks 8–10 are in place at G0; (b) the non-contact sensor mounts above an existing conveyor with read-only data, so installation takes days; (c) specimens come from the host's existing shift sampling and belt cuts, so 70 microscope specimens and 44 belt composites (each split for Bond and mineralogy references) over about 16 weeks need only about 5 and 3 a week; (d) samples are dispatched weekly and results return within an agreed three weeks of dispatch; and (e) the software and abundance checks already exist, so month 1 goes on local data and development-only method selection, with calibration fitted later on its own cohort. No host, laboratory or sensor access is arranged yet, and ≤10% remains a target. Two limits apply: the sensor is mounted only under the bench outcome matrix (Table 13, week 8), and the four-week advisory window can show the workflow, timing and operator response, but cannot measure a gain of about 2% with useful precision, so independently measured value (G5) needs a longer trial after the pilot. If supply, delivery or the plant stop slips, the pilot is extended; the cohort is never cut to keep the date. The added mineral work must be reflected at G0 in quotes, sample access, laboratory throughput and staff capacity.

7.2 Pilot objectives and acceptance targets

Table 7 sets out the proposed targets, signed with the host before any evaluation specimen is collected. We use "90% accuracy" in one strict sense for the microscope: at least 90% of locked specimens must meet the error contract for every decision-critical phase. With 30 locked specimens, an observed 27 of 30 shows only that the true pass rate exceeds 76% (one-sided 95%); under independent sampling, outcomes such as 29/29 or 45/46 can place the one-sided 95% lower bound

Table 6: Proposed mineral decision; "host" marks cells to confirm at P0.

Stream, owner	UG2 or Merensky; host metallurgist (to appoint)
Endpoint	Phases named by host mineralogist (candidates: sulphides; chromite for UG2 [6])
Action, actuator	Investigation alert first; blend or hold only via existing stockpile, blending point or manual procedure (host)
Expiry, check	Expires at material arrival; mineralogy on matched cut
Outcome	First: detection lead time, false-alert burden, usefulness; avoided off-spec feed only where a host intervention is executed

above 90%, but these are illustrative confidence calculations, not a powered design (at a true 95% pass rate, 29/29 occurs only about 23% of the time); the final design accounts for expected pass rate, power, parcel dependence and a fixed stopping rule. So 27 of 30 is a proposed observed acceptance rule, not proof of a 90% pass rate; 46 specimens would add about R0.56 million before contingency. On the belt, 24 composites give exploratory evidence of limited precision (22 of 24 within the contract bounds the pass rate only above 76%), and the exact pass-count rule, handling of every critical phase, refusals and reporting limits are fixed before testing; the sampled cut must match the scanned parcel although the camera sees only the surface, and microscopy acceptance counts are never reused as belt evidence; and because a bound can cover 90% simply by being too wide, its width and refusal rate are judged with its coverage. Before any actuation, each adjustment follows measured input, supported estimate, independently calibrated plant-response relation, constrained candidate action, operator approval and observed outcome reconciled with the laboratory; error alone does not set a safe or profitable setting. *Proposed mineral decision (hypothesis to validate).* Table 6 sets out one candidate decision; P0 selects decision-critical phases with the host and laboratory and verifies that the sensor range and acquisition conditions carry enough information for them (candidates are not claims of SWIR capability; any change of phase set is declared before evaluation), and chooses a single primary action and confirms the equipment and authority that make it possible. The funded camera cannot hold or redirect ore, an investigation alert is not a conveyor stop, sulphide abundance is not treated as PGM grade, and mineral benefits are kept separate from the hardness throughput simulation.

The pilot then measures whether estimates improve that decision within its deadline, net of operating, verification and error costs; fewer laboratory submissions would be a later, separately validated policy.

7.3 Work plan and daily site routine

Figure 4 illustrates the plan, highlighting that the sensor is tested on host ore before it is mounted, and that the microscope model is frozen before any locked reference is revealed. A typical day on site (proposed; the host sets hours and safety rules) runs: (i) handover, safety talk and sensor checks (references, window, purge air, overnight refusals); (ii) collecting sampler and belt cuts time-stamped to their parcels, logged in the custody register and split for sections, Bond, mineralogy and retention, with the custodian's script assigning each to development, calibration or locked; (iii) imaging returned sections under controlled light, with locked outputs sealed before any reference is seen; (iv) development work on development data only, while the custodian stores locked shadow references without releasing per-specimen results until the campaign and prediction seal are complete, and, in the advisory window, reviewing each proposal with the operator; and (v) a review with the host metallurgist and signed backups. A reference batch goes out weekly, and a monthly steering meeting checks spend, risks and schedule.

Table 7: Proposed pilot targets. "Agreed" values are signed with the host and statistician before evaluation data are collected.

Target	Pass condition (proposed)	Gate
Phase abundance	Pentlandite, chalcopyrite, pyrrhotite: $\leq$10% relative error above an agreed reporting limit, absolute tolerance below it; observed rule $\geq$27 of 30 locked specimens (not proof of a 90% population rate)	G1A
Coverage	Refusals and failures stay in the denominator; answered share $\geq$ agreed minimum	G1A
Bench hardness (readiness screen)	20 host composites, split before collection into development and calibration subsets; exploratory if calibration informs fitting; recipe frozen before shadow evaluation: error below a metadata baseline; agreed bound width	G1B
Bench minerals (readiness screen)	Same split, with laboratory mineralogy: decision-critical phases beat a metadata baseline on an agreed basis and reporting limit	G1M
Belt shadow	24 belt composites, locked evaluation of a frozen version, no online recalibration. Hardness: predefined count inside the bound, width, refusals. Minerals: $\leq$10% relative error per critical phase above the reporting limit, coverage, grouping by sampling event. End-to-end latency (p50/p95/p99) inside the transit time	G3
Advisory window	Proposals withdrawn when they should be; no setpoint outside the envelope; operator response logged	G4

7.4 Pilot team

Table 8 lists the people the pilot needs. Lethabo Hoaeane and Sibusiso Khumalo jointly lead the pilot with equal decision-making authority, each accountable for their assigned technical workstream (allocation proposed, pending both co-leads' acceptance). Material changes to scope, budget, milestones, external commitments, model promotion or release need both; neither has a casting vote, and a disagreement pauses the change for specialist or steering advice. The co-leads coordinate evaluation but are not its independent approvers and have no access to locked labels. The software and operational-technology (OT) security engineer must be recruited before G0, and the independent roles are contracted, so that assessment never collapses into self-certification. The host owns plant operation and stop authority; in shadow nothing is written to the plant, and in the advisory window only an operator can change a setpoint.

7.5 Funding structure and the role of TIA

TIA funds technology from proof of concept to pre-commercialisation [31]. Its Technology Development Fund (TRL 4–7; demonstration and pilot plants) is the route we propose to investigate for the laboratory track; the belt components are self-assessed at TRL 3 or below, so their eligibility is checked in P0 before any instrument is allocated. The Seed Fund is listed for TRL 3–8 [31]; one SAMRC-run call (TRL 5–6, up to R1.5 million) shows what an application needs, a prototype, roadmap, industry partner and IP strategy [32], not our eligibility, and the late-stage Industry Matching Fund is not assumed. Terms are confirmed with TIA and a technology-

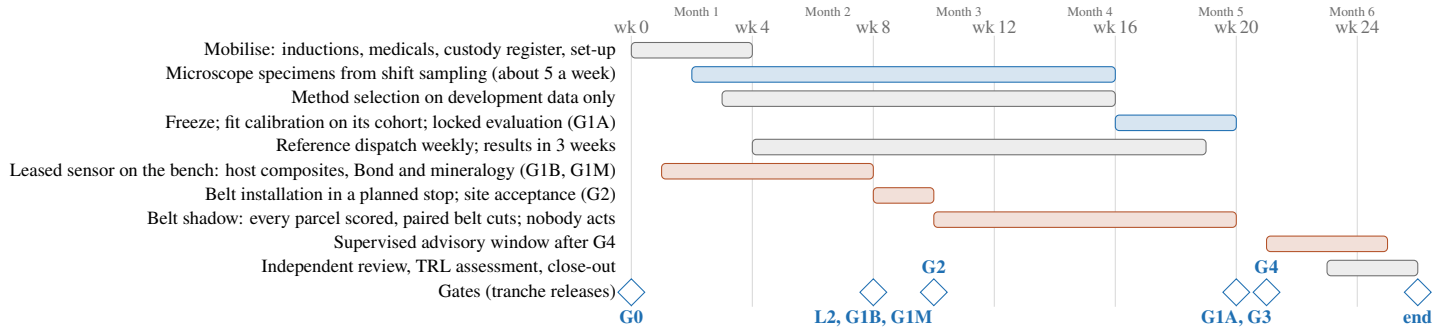


Figure 4: The six-month pilot, in weeks from the readiness event (planning estimate). Blue bars: microscope track; copper bars: belt track. G0 releases L1 and B1; the week-8 laboratory review releases L2; the bench outcome matrix (Table 13) decides B2, R-H and R-M (Table 4). G1A falls in week 20 with G3. If both bench gates fail, the leased sensor is returned before installation and the microscope track continues.

Table 8: The pilot team (functions; one person may cover several, except the independent ones).

Role	Time	During the pilot
Co-lead L. Hoaeane: belt models, evaluation	Full time, on site	Belt models (hardness, minerals), economics; shared planning, budget, host, gates
Co-lead S. Khumalo: microscopy, field, systems	Full time, on site	Microscope, imaging, integration, verification, reporting; shared planning, budget, host, gates
Software and OT-security engineer	To recruit	Sensor integration, offline operation, signed record
Independent mineralogist	35 days	Development annotation, reference review
Statistician and custodian	18 days	Blind split, sealed predictions, evaluation
Vendor engineer	Belt allowance	Installation, calibration, acceptance
Host metallurgist, technicians, operators, safety and OT	In kind	Use case, sampling, envelope, lockout, sign-off

Table 9: Proposed funding structure around TIA (proposals; nothing is agreed).

Stage	TRL	TIA instrument	Cash and other parties
Pilot: laboratory, team, references	Subject to assessment	Technology Development Fund (eligibility unconfirmed)	TIA R5.59 M; host pays Bond tests (R1.00 M with contingency) and gives site access, staff time and laboratory space; Mintek reference work at cost
Pilot: belt	Subject to assessment	Eligibility unconfirmed; checked before G0	R4.10–6.56 M; team proposes 50:50 with the host (R2.05–3.28 M each), not an agency term
Microscope product	5–6	Seed Fund, terms of the current call	To be confirmed; technology-transfer office
Value trial and roll-out	7–9	Matching or Commercialisation Support	Host-led; investors

transfer office; Table 9 is a team proposal, with nothing applied for or agreed.

7.6 Pilot risks and mitigations

Table 10 sets out the risks of the pilot period, as the team assesses them, each with a mitigation before it happens and a fallback if it does. Contingency pays only for uncertain costs inside the approved scope. Three fallbacks run through all of them: the host's controls keep authority, each stage ends at a gate where spending can stop, and no success fee is due without measured value.

7.7 Governance, compliance and training

These are planned controls, not achieved compliance; a qualified review of applicability and cited versions is obtained before submission. The Mine Health and Safety Act requires a risk assessment with the health and safety committee, and training before significant changes to plant [34], which applies to the belt installation; POPIA governs approvers' names in the record [35]; Mining Charter clause 2.3.1 counts skills and R&D spend [36]; IP from public funding falls under the IPR Act [37], managed through MOTT under the hackathon agreement [8], although the contract settles ownership; and Mintek and TIA spend under the PFMA [38]. Independent checks would be commissioned, subject to agreement and method capability, from a laboratory accredited by SANAS for the selected method's scope [39], Mintek's QEMSCAN, a value auditor using a plan adapted from IPMVP [30], an IEC 62443 assessor, an IP attorney through MOTT and a university model reviewer. Each group is trained before its phase, aligned with the site's skills plan [40]: laboratory staff and metallurgists before G1A work starts, instrument technicians at installation, and control-room operators before G4; competence is shown by observed tasks and sign-off, not attendance.

8 TRL milestones and gates

Table 11 places each component on the standard readiness scale [1] as a provisional self-assessment. Passing a readiness screen or beating a weak baseline does not by itself earn a level; the jumps in Table 11 need an integrated demonstration and assessor review. A level is assessed by an appointed independent reviewer from demonstrated performance against agreed criteria, not awarded automatically at a gate or from elapsed time. The belt system climbs furthest in the pilot, because it moves from public core to the host's own belt in two steps: host samples on the bench (TRL 4, G1B for hardness, G1M for minerals) and shadow operation on the belt (TRL 6, G3). The advisory window starts operational evidence towards TRL 7; independently measured financial value (G5) is a separate commercial gate, not a TRL criterion.

Table 12 lists what each gate must show, who decides, and what happens if it fails. After the pilot, an extended advisory trial with on/off operating blocks gives the independently measured value for G5, so a commercial decision is about 12

Table 10: Pilot-period risk register: mitigation before the event, and the fallback if it happens.

#	Risk during the pilot	Mitigation (before)	Fallback (if it happens)
P1	Fewer than 5 specimens a week, laboratory turnaround over 3 weeks, or quotes above budget	Sampling plan and turnaround in quotes at G0; second laboratory pre-qualified	Extend collection; narrow or seek a new decision; never shrink the locked cohort
P2	Sensor late, no lease unit, or plant stop missed for installation	Delivery, lease and stop booked at G0; mount tested on the bench	Shift the belt track; shorten the advisory window, not shadow
P3	One or both bench gates fail (G1B, G1M)	Bench tests on host composites before mounting	Apply the bench matrix; if both fail, do not release B2, settle B1 or cancellation; bench references already paid stay with the laboratory track
P4	Belt drift (dust, moisture, wavelength) or scan–sample timing mismatch	Enclosure, purge air, references every cycle; time-stamped parcels	Refuse affected parcels; exclude unmatched pairs and report them
P5	Operator over-trust, unsafe write or cyber incident	Deadband; training before G4; read-only shadow; advisory tags only; IEC 62443 zones [33]	Advice expires or is disconnected; the plant runs on its own controls
P6	The microscope misses the $\leq$10% target, or data leak between sets	One frozen model; independent custodian; sealed predictions	Report the negative result; collect a fresh cohort before any new claim
P7	Engineer not recruited, host withdraws, or data and IP terms unsigned	Recruit and sign before G0; named deputies; LumenStone rights clarified	Hold the next tranche; keep lawful evidence; qualify another host
P8	Belt surface spectra do not represent the parcel's bulk mineral composition	Cuts matched to scanned parcels; grouping by sampling event	Report the negative result; drop the mineral endpoint; hardness continues only if its own gates and approvals pass
P9	Rand 10% weaker on US$-priced lines; safety incident at the belt	Rand quotes where possible; lockout, escort and permits near moving plant	About R0.64–0.89 M, met from contingency; stop work and follow host procedure

Table 11: Provisional self-assessed TRL now, and the evidence that could support each next level.

Component	Now	Evidence for assessment toward, if criteria are met	After
Microscope workflow	Provisional laboratory proof of concept; advice unvalidated	5: independent host specimens (G1A); 6: shift routine	7–8
Belt hardness model	3: proof of concept on public core; simulated plant	4: host samples on the bench (G1B); 6: shadow on the host belt (G3)	7
Belt mineral model	Below 3: exploratory public analogue data, no PGM belt validation; does not beat a metadata baseline	4: bench (G1M); 6: shadow (G3)	7
Decision layer and record	4: built; replayed data; simulated plant	6: shadow; 7 begun in the advisory window (G4)	7–8

months from the readiness event.

Public paired data and planned validation. The product's output is a time-stamped parcel estimate with uncertainty and a valid/invalid measurement status. Public paired MLA–hyperspectral data (204 sections, 49 drillholes, seven locations [10]) have been inspected and used for exploratory site-held-out development experiments. A 20-specimen alignment sensitivity comparison did not materially improve cross-site prediction. Specimen bG11A5 is on hold for unresolved reference geometry (exclusion would mean refitting affected folds). Source audits found malformed mask exports, uncertain sensor alignment and label-provenance differences; reproducing export stages (and one full chain) shows computational reproducibility, not physical registration. Infrared fusion awaits registration and calibration evidence, drillhole grouping needs a verified crosswalk, and these sections say nothing about moving PGM ore. None of this replaces matched PGM data.

Table 12: Gates, decision owners and stop outcomes. Numerical thresholds are signed before evaluation data are collected.

Gate	Release condition	Decides	If it fails
G0	Host letter, quotes, sensor lease and delivery, plant stop booked, data and IP terms, signed contract	Funder, host	Preparation only
L2	Laboratory review (week 8): agreed counts collected and split, reference turnaround within the agreed limit, discrepancies resolved, quotes accepted (values set by agreement)	Co-leads, funder, custodian	Hold L2; fix or narrow
G1B	Hardness transfers to host composites on the bench (week 8)	Reviewer, host	Table 13
G1M	Mineral estimates beat a metadata baseline on bench composites (week 8; exploratory readiness screen, not target validation)	Reviewer, mineralogist	Table 13
G2	Risk assessment, read-only design, site acceptance (week 10)	Host OT, safety	Repair or remove
G1A	Microscope targets met on locked specimens (week 20)	Reviewer, host	Image review only
G3	Belt shadow targets for hardness and minerals: coverage, timing, drift, refusals (week 20)	Metallurgist	Extend or stop
G4	Envelope, change management, training, rollback (week 21)	Change board	No advisory window
G5	Independently measured net value and total cost (after the pilot)	Auditor, host	No success fee

9 Conclusions and recommendation

The present contribution is a functioning, inspectable evidence workflow and a falsifiable route to local validation. On public analogue data the hardness model improves on simple baselines and the microscope classifies three relevant sulphide phases, but the feed-rate benefit is a simulation, the microscope results are development-influenced, the active application checkpoint has shown large abundance errors on one section, and the $\leq$10% error target is untested. The case is for bounded feasibility, not guaranteed target achievement or automatic TRL promotion. We recommend funding the bounded P0 feasibility stage first. Progression to the proposed six-month programme requires host access, scoped quotations, an agreed measurement and decision contract, and the staged re-

Table 13: Bench outcome matrix (week 8). A minerals-only path cannot claim the hardness-derived 1.7–1.9% throughput simulation.

G1B	G1M	Proposed outcome
Pass	Pass	Both endpoints proceed, subject to G2 and release approval
Pass	Fail	Hardness only; mineral claims and unneeded R-M orders removed
Fail	Pass	Minerals only, after a mineral-driven decision, compatible hardware and a revised value and acceptance plan are approved; otherwise shadow only
Fail	Fail	No installation; leased equipment returned under contract; laboratory track judged separately

lease approvals in Section 7. The ≤10% local target remains untested; commercial claims require independent reference agreement, useful coverage and measured net value on the affected stream.

CRediT authorship contribution statement. **Lethabo Hoaeane:** conceptualisation, belt hyperspectral modelling, belt monitor and decision application, economics, presentation. **Sibusiso Khumalo:** microscope model and evaluation, advisor and safety gates, OPC UA simulation, verification, technical report.

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Use of AI tools. AI assistants helped with code and drafting; results link to scripts and evidence in the manifest, planning allowances are labelled, and AI checks do not replace specialist validation.

Data and code availability. Dataset licences are given in Section 2. Public revision: https://github.com/Sibusiso-K/KHANYA; later evidence is local and manifest-bound (the supporting evidence note). Research application (account required): https://reefprint-judges.ambitiousflower-ff5d4625.francecentral.azurecontainerapps.io/; the older static Hugging Face replay is a different build.

Supporting evidence. *Evidence:* https://github.com/Sibusiso-K/KHANYA/releases/tag/report-evidence-v20 (ZIP) and `report/EVIDENCE_MANIFEST.md` (ledger, replay, hashes, release index, full B02/B04 counts). *B02* (copper development data): PLS did not beat a metadata lookup (e.g. chalcopyrite 0.28 vs 0.17 points; 21/65 within 10% on unseen lines). *Microscope* (historical checkpoint `de7135a9`; recall/precision): chalcopyrite 0.716/0.746, pyrrhotite 0.934/0.926, pentlandite 0.742/0.675. Evidence is frozen at the 11 October 2026 handover; the manifest lists every included run (EXP21–26, revision 0000007 and image digest); revision 0000008 is a separately hashed frontend addendum, and later work is excluded. *B03:* the tested local v3.1 build rejects altered intake and recomputes saved estimates; software consistency only.

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All URLs were opened on 5, 7 or 8 October 2026.

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